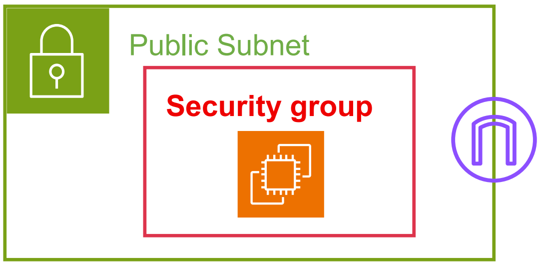
AWS Solution Architect Training with
AWS Cloud Practitioner Global Certification

Capstone Projects

Trainer: Aravindraj.G- Nminds Academy

AWS Academy Certified Educator

Secure the Custom VPC using Public Subnet in AWS



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Sri Eshwar College of Engineering,

Coimbatore.

Objective

A **VPC (Virtual Private Cloud)** in Amazon Web Services (AWS) is a **logically isolated virtual network** inside AWS where you can launch and manage your cloud resources securely.

- Amazon Virtual Private Cloud (**VPC**) is a service that lets you launch **AWS** resources in a logically isolated virtual network that you define.
- It is logically isolated from other virtual networks in the AWS Cloud.
- A virtual private cloud (VPC) is a secure, isolated private cloud hosted within a public cloud.
- Amazon Virtual Private Cloud (Amazon VPC) hides the complexity, and simplifies the deployment of secure private networks.

Subnets:

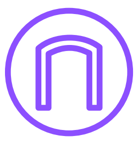
A subnet is a sub range of IP addresses in the VPC. AWS resources can be launched into a specified subnet.

- After creating a VPC, you can add one or more subnets in each Availability Zone.

- ✓ *The public subnet* resources that must be connected to the internet.
- ✓ The *private subnet* resources that must remain isolated from the internet.

Internet Gateway:

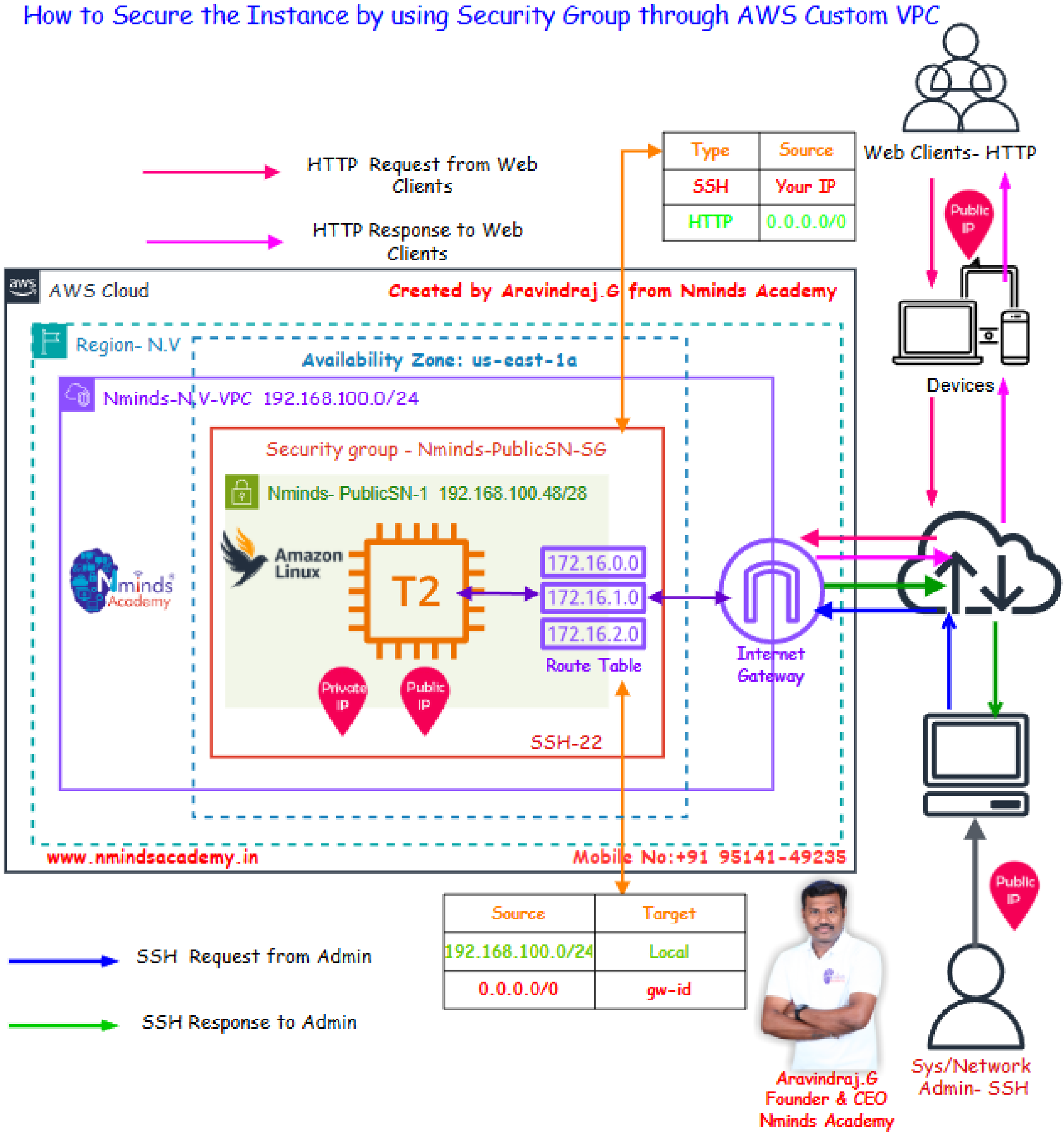
An *internet gateway* is a VPC component that allows communication between resources in your VPC and the internet. It is horizontally scaled, redundant, and highly available. An internet gateway supports IPv4 and IPv6 traffic. An internet gateway serves two purposes. First, it provides a target in your VPC route tables for internet-routable traffic. Second, the internet gateway performs network address translation (NAT) for instances that were assigned public IPv4 addresses.



Internet gateway

Topology

How to Secure the Instance by using Security Group through AWS Custom VPC



Used Services in AWS

Amazon VPC

Amazon Elastic Compute Cloud (Amazon EC2)

Used Services in Windows Server

APACHE HTTP SERVER

Apache Webserver

Public subnet

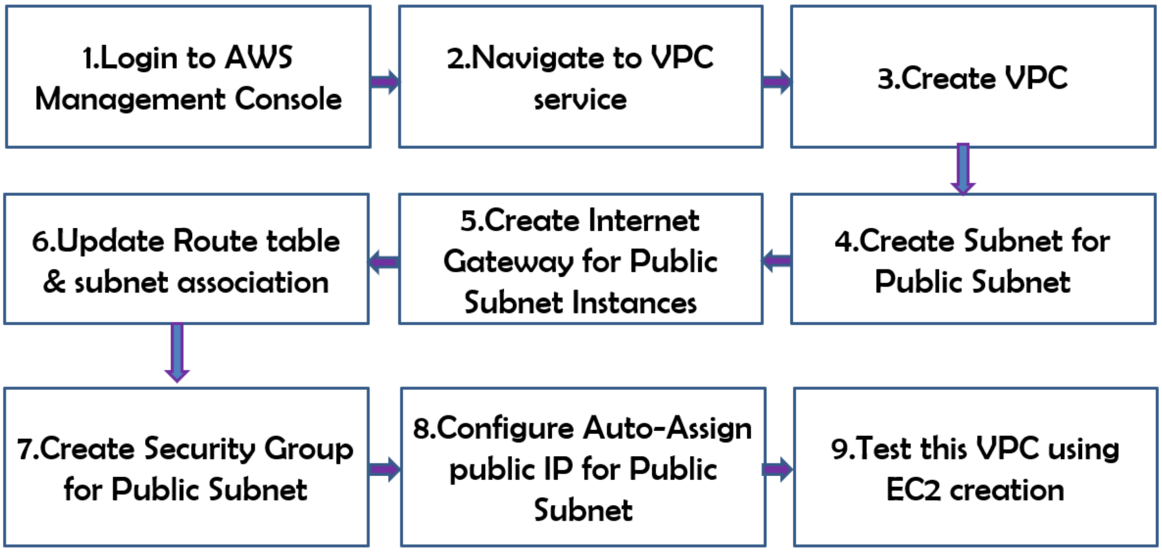
Internet gateway

172.16.0.0
172.16.1.0
172.16.2.0

Route table

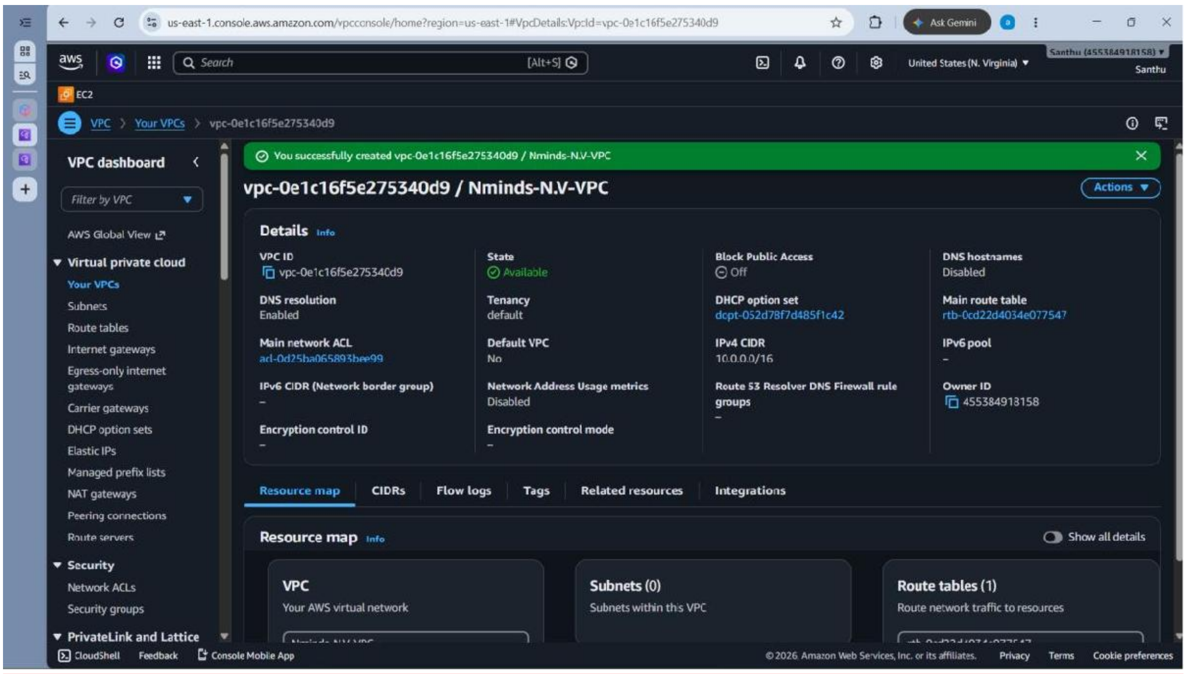
Execution Steps:

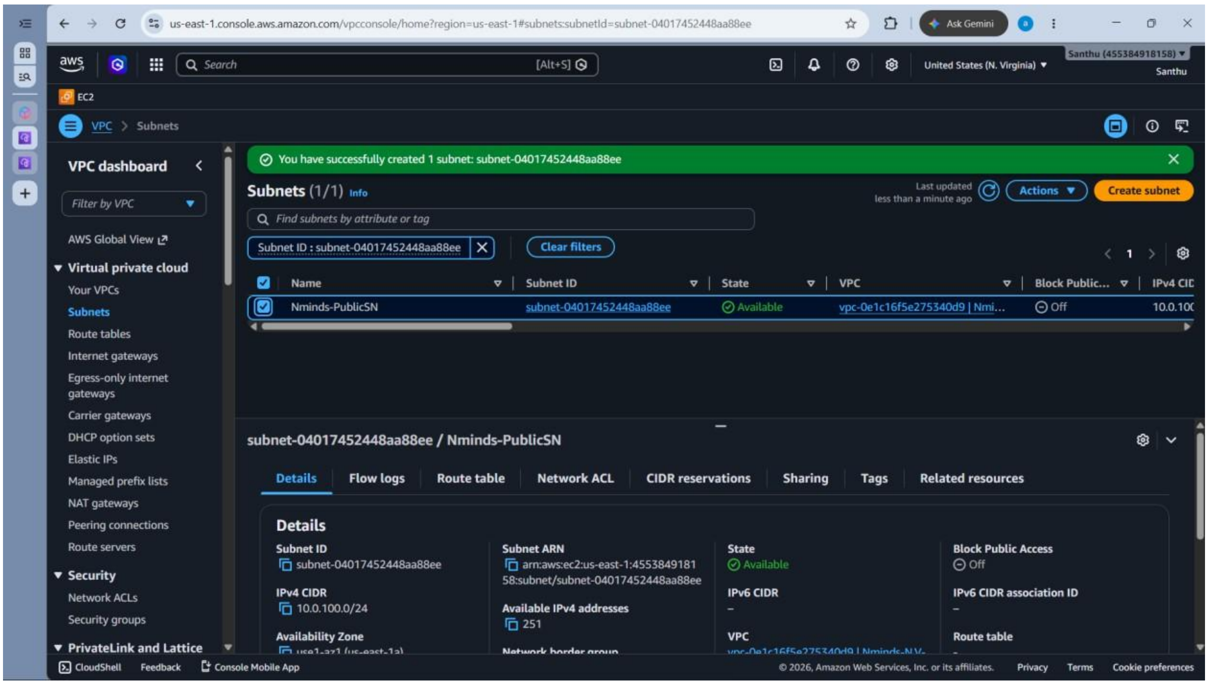
Steps to Configure Custom VPC in AWS

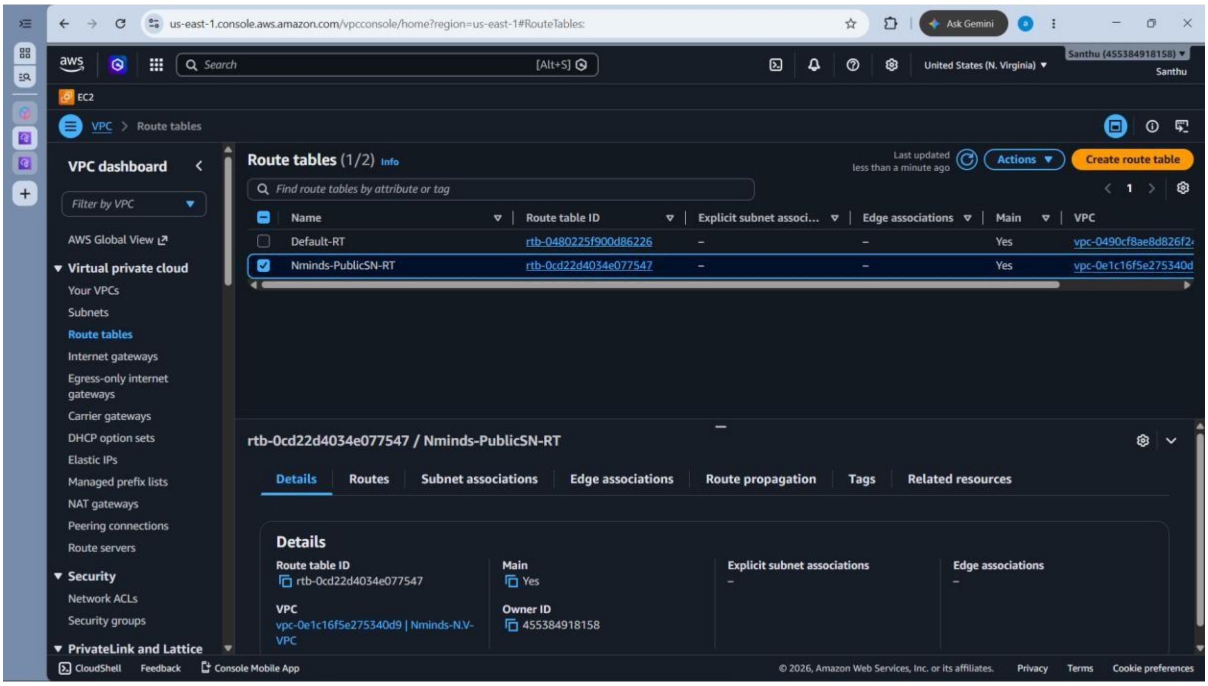


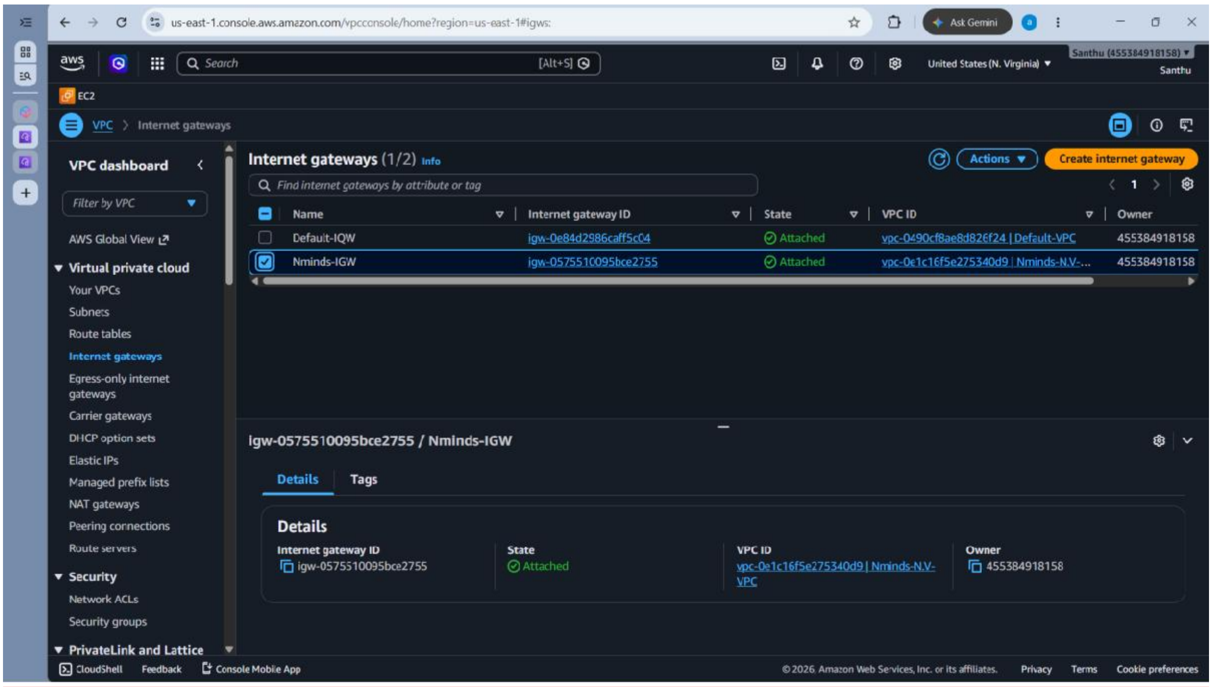
Execution Tasks:

Task1: Create the custom VPC with Public Subnet based on the above topology:




us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#subnets:subnetId=subnet-04017452448aa88ee
VPC > Subnets
VPC dashboard
You have successfully created 1 subnet: subnet-04017452448aa88ee
Subnets (1/1)
Actions
Create subnet
Subnet ID : subnet-04017452448aa88ee
Clear filters
Name | Subnet ID | State | VPC | Block Public... | IPv4 CIDR
Nminds-PublicSN | subnet-04017452448aa88ee | Available | vpc-0e1c16f5e275340d9 | Nmi... | Off
subnet-04017452448aa88ee / Nminds-PublicSN
Details
Flow logs
Route table
Network ACL
CIDR reservations
Sharing
Tags
Related resources
Subnet ID
subnet-04017452448aa88ee
Subnet ARN
arn:aws:ec2:us-east-1:455384918158:subnet/subnet-04017452448aa88ee
State
Available
Block Public Access
Off
IPv4 CIDR
10.0.100.0/24
Available IPv4 addresses
251
IPv6 CIDR
IPv6 CIDR association ID
Availability Zone
VPC
Route table
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
PrivateLink and Lattice


us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#RouteTables:
VPC > Route tables
VPC dashboard
Route tables (1/2)
Actions
Create route table
Name | Route table ID | Explicit subnet associ... | Edge associations | Main | VPC
Default-RT | rtb-0480225f900d86226 | – | – | Yes | vpc-0490cf8ae8d826f2
Nminds-PublicSN-RT | rtb-0cd22d4034e077547 | – | – | Yes | vpc-0e1c16f5e275340d
rtb-0cd22d4034e077547 / Nminds-PublicSN-RT
Details
Routes
Subnet associations
Edge associations
Route propagation
Tags
Related resources
Route table ID
rtb-0cd22d4034e077547
Main
Yes
Explicit subnet associations
Edge associations
VPC
vpc-0e1c16f5e275340d9 | Nminds-N.V-VPC
Owner ID
455384918158


us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#igws:
VPC > Internet gateways
VPC dashboard
Filter by VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
PrivateLink and Lattice
Internet gateways (1/2) Info
Actions
Create internet gateway
Find internet gateways by attribute or tag
Name | Internet gateway ID | State | VPC ID | Owner
Default-IGW | igw-0e84d2986caff5c04 | Attached | vpc-0490cf8ae8d826f24 | Default-VPC | 455384918158
Nminds-IGW | igw-0575510095bce2755 | Attached | vpc-0e1c16f5e275340d9 | Nminds-N.V-... | 455384918158
igw-0575510095bce2755 / Nminds-IGW
Details
Tags
Details
Internet gateway ID
igw-0575510095bce2755
State
Attached
VPC ID
vpc-0e1c16f5e275340d9 | Nminds-N.V-VPC
Owner
455384918158
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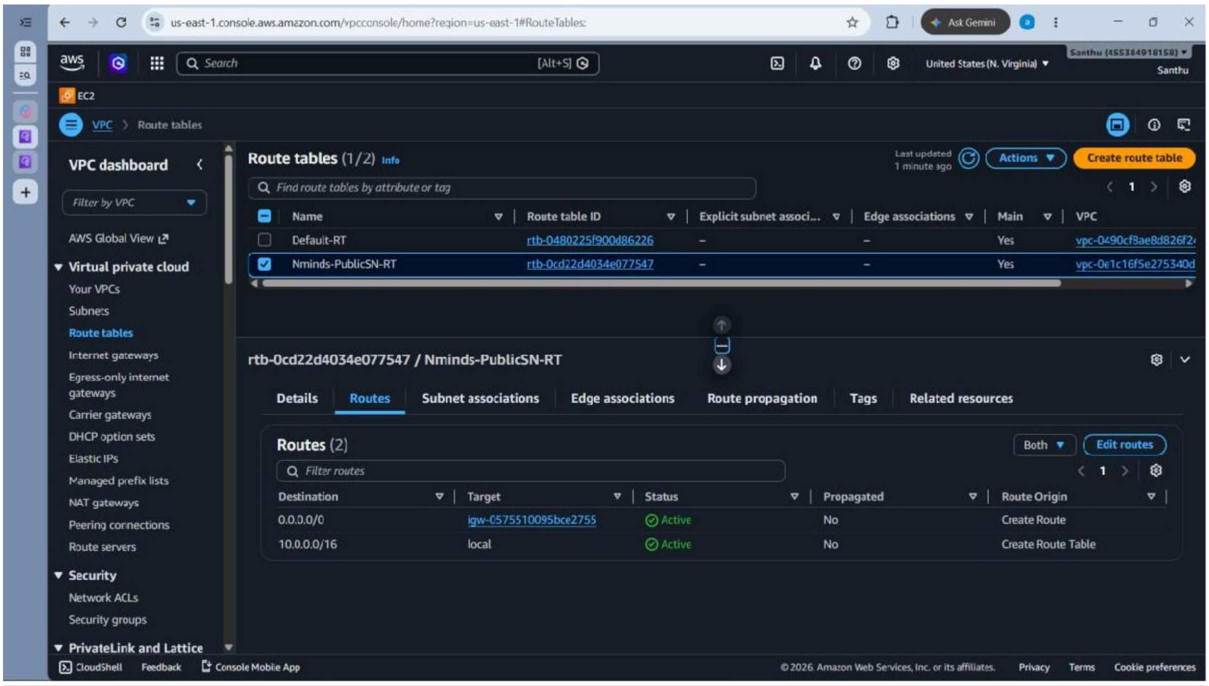
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#RouteTables:
VPC > Route tables
VPC dashboard
Route tables (1/2) Info
Last updated 1 minute ago
Actions
Create route table
Find route tables by attribute or tag
Name | Route table ID | Explicit subnet associ... | Edge associations | Main | VPC
Default-RT | rtb-0480225f900d86226 | – | – | Yes | vpc-0490cf8ae8d826f2
Nminds-PublicSN-RT | rtb-0cd22d4034e077547 | – | – | Yes | vpc-0e1c16f5e275340d
rtb-0cd22d4034e077547 / Nminds-PublicSN-RT
Details
Routes
Subnet associations
Edge associations
Route propagation
Tags
Related resources
Routes (2)
Both
Edit routes
Filter routes
Destination | Target | Status | Propagated | Route Origin
0.0.0.0/0 | igw-0575510095bce2755 | Active | No | Create Route
10.0.0.0/16 | local | Active | No | Create Route Table
CloudShell Feedback Console Mobile App
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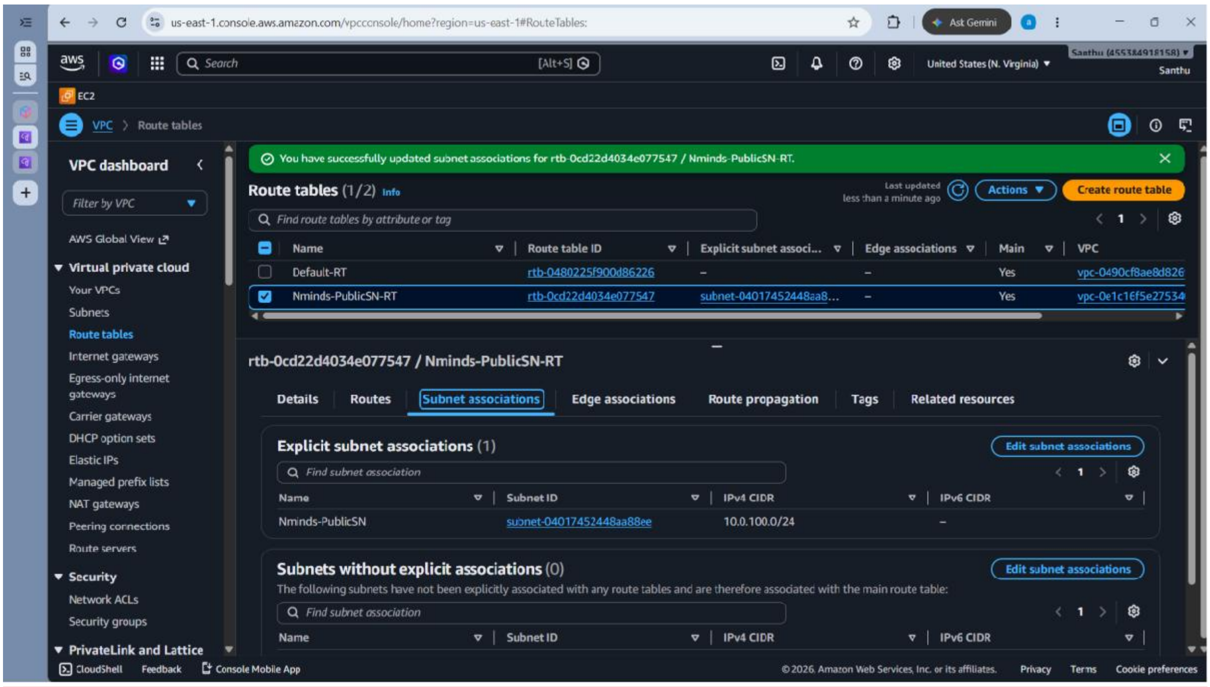
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#RouteTables:
Ask Gemini
Search
[Alt+S]
United States (N. Virginia)
Santhu
EC2
VPC > Route tables
VPC dashboard
Filter by VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
PrivateLink and Lattice
You have successfully updated subnet associations for rtb-0cd22d4034e077547 / Nminds-PublicSN-RT.
Route tables (1/2) Info
Last updated less than a minute ago
Actions
Create route table
Find route tables by attribute or tag
Name
Route table ID
Explicit subnet associ...
Edge associations
Main
VPC
Default-RT
rtb-0480225f900d86226
Yes
vpc-0490cf8ae8d826
Nminds-PublicSN-RT
rtb-0cd22d4034e077547
subnet-04017452448aa8...
Yes
vpc-0e1c1ef5e27534
rtb-0cd22d4034e077547 / Nminds-PublicSN-RT
Details
Routes
Subnet associations
Edge associations
Route propagation
Tags
Related resources
Explicit subnet associations (1)
Edit subnet associations
Find subnet association
Name
Subnet ID
IPv4 CIDR
IPv6 CIDR
Nminds-PublicSN
subnet-04017452448aa88ee
10.0.100.0/24
Subnets without explicit associations (0)
Edit subnet associations
The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table:
Find subnet association
Name
Subnet ID
IPv4 CIDR
IPv6 CIDR
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Feedback
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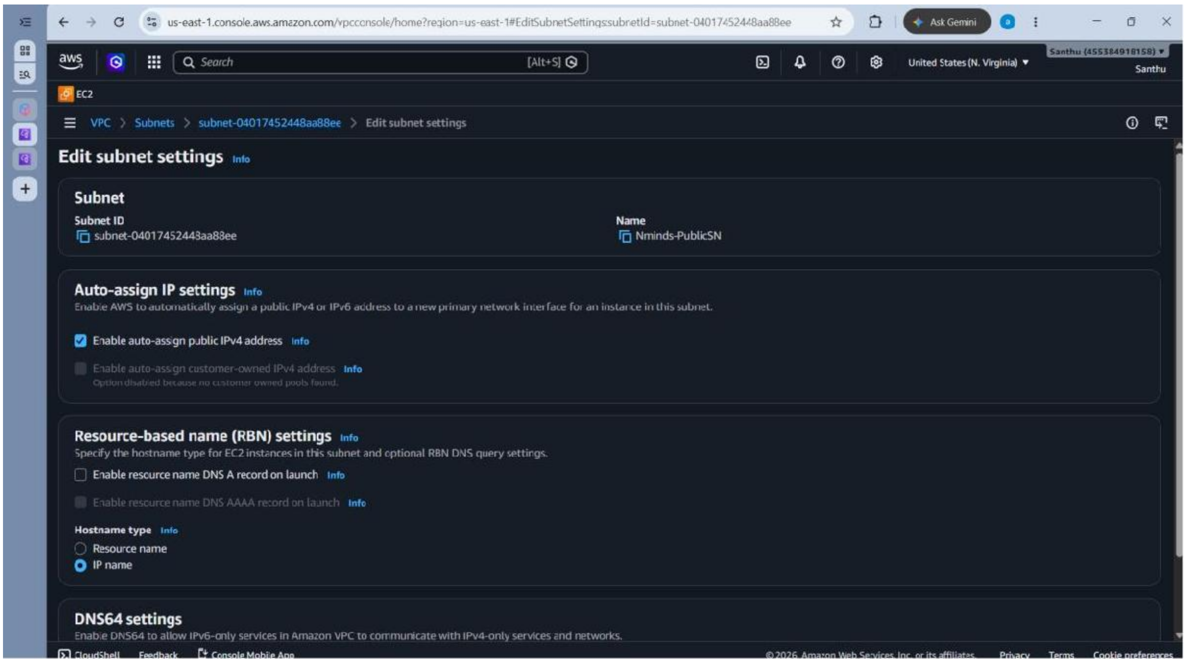
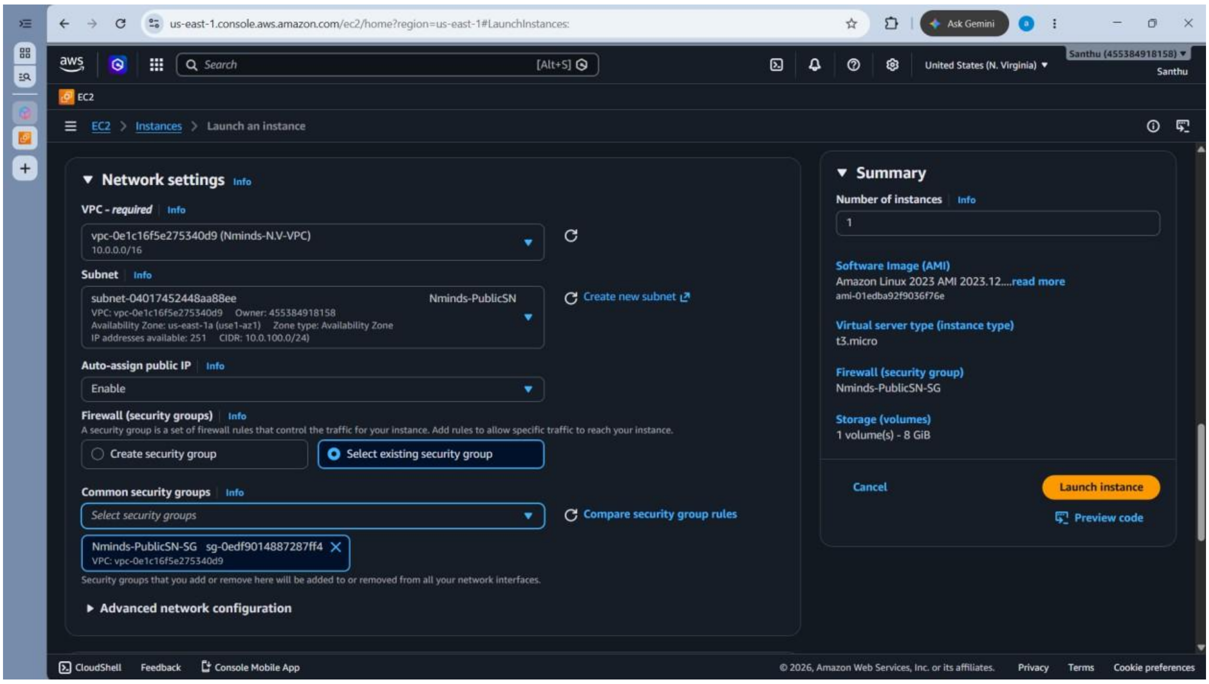
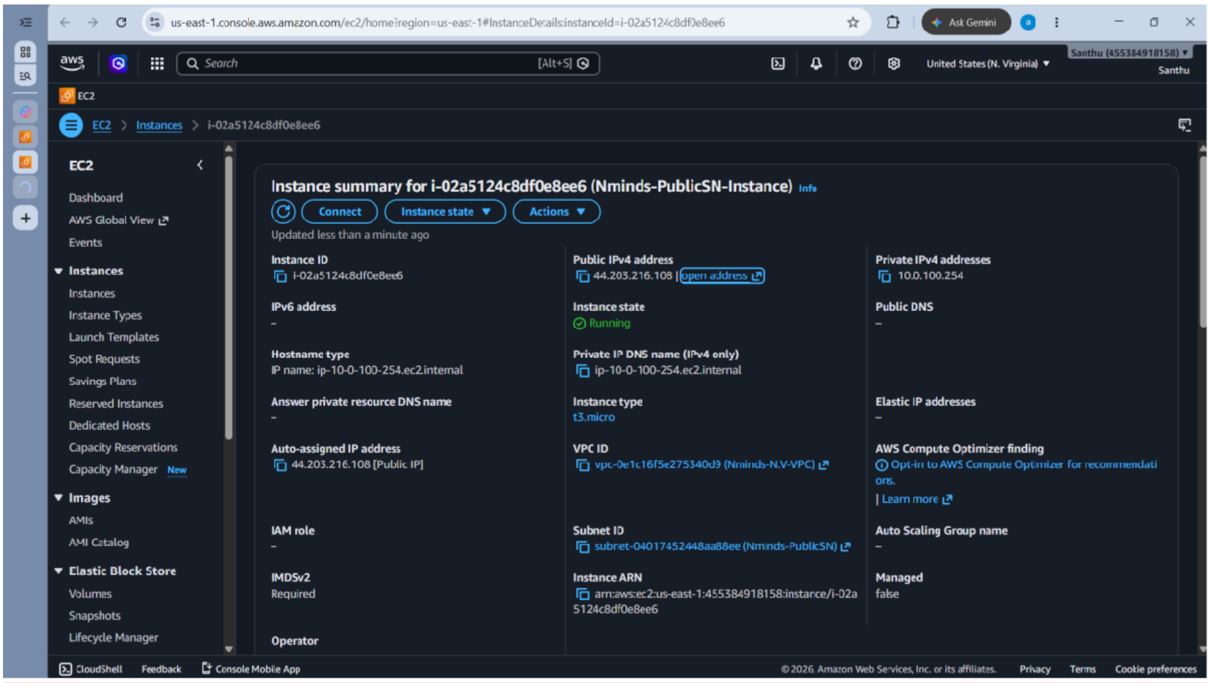
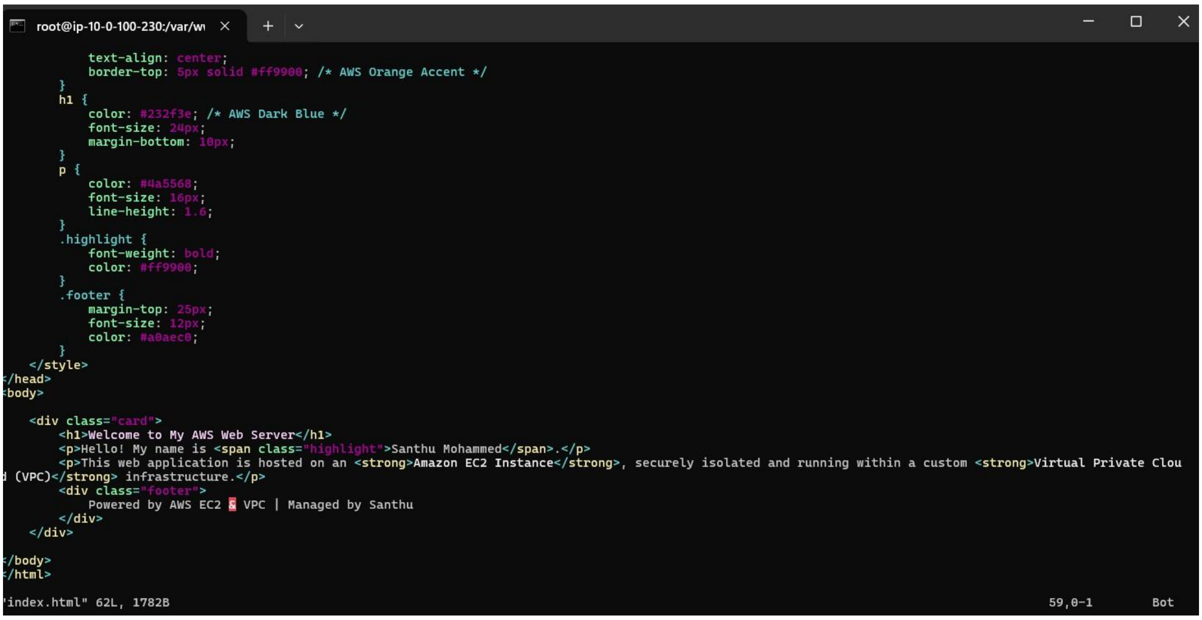
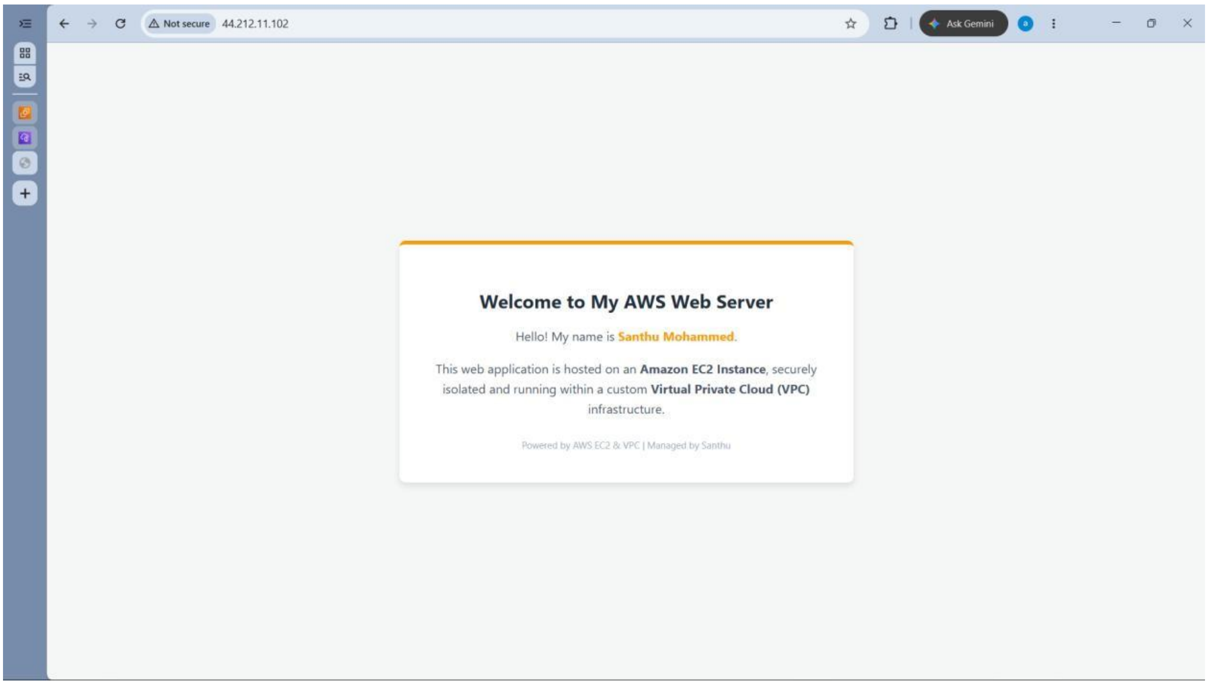

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#EditSubnetSettings:subnetId=subnet-04017452448aa88ee
Ask Gemini
Search
[Alt+S]
United States (N. Virginia)
Santhu
EC2
VPC > Subnets > subnet-04017452448aa88ee > Edit subnet settings
Edit subnet settings Info
Subnet
Subnet ID
subnet-04017452443aa88ee
Name
Nminds-PublicSN
Auto-assign IP settings Info
Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.
Enable auto-assign public IPv4 address Info
Enable auto-assign customer-owned IPv4 address Info
Option disabled because no customer owned pools found.
Resource-based name (RBN) settings Info
Specify the hostname type for EC2 instances in this subnet and optional RBN DNS query settings.
Enable resource name DNS A record on launch Info
Enable resource name DNS AAAA record on launch Info
Hostname type Info
Resource name
IP name
DNS64 settings
Enable DNS64 to allow IPv6-only services in Amazon VPC to communicate with IPv4-only services and networks.
CloudShell
Feedback
Console Mobile App


us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:
EC2 > Instances > Launch an instance
Network settings Info
VPC - required Info
vpc-0e1c16f5e275340d9 (Nminds-N.V-VPC)
10.0.0.0/16
Subnet Info
subnet-04017452448aa88ee Nminds-PublicSN
VPC: vpc-0e1c16f5e275340d9 Owner: 455384918158
Availability Zone: us-east-1a (use1-az1) Zone type: Availability Zone
IP addresses available: 251 CIDR: 10.0.100.0/24)
Create new subnet
Auto-assign public IP Info
Enable
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
Common security groups Info
Select security groups
Compare security group rules
Nminds-PublicSN-SG sg-0edf9014887287ff4
VPC: vpc-0e1c16f5e275340d9
Security groups that you add or remove here will be added to or removed from all your network interfaces.
Advanced network configuration
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....read more
ami-01edba92f9036f76e
Virtual server type (instance type)
t3.micro
Firewall (security group)
Nminds-PublicSN-SG
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code

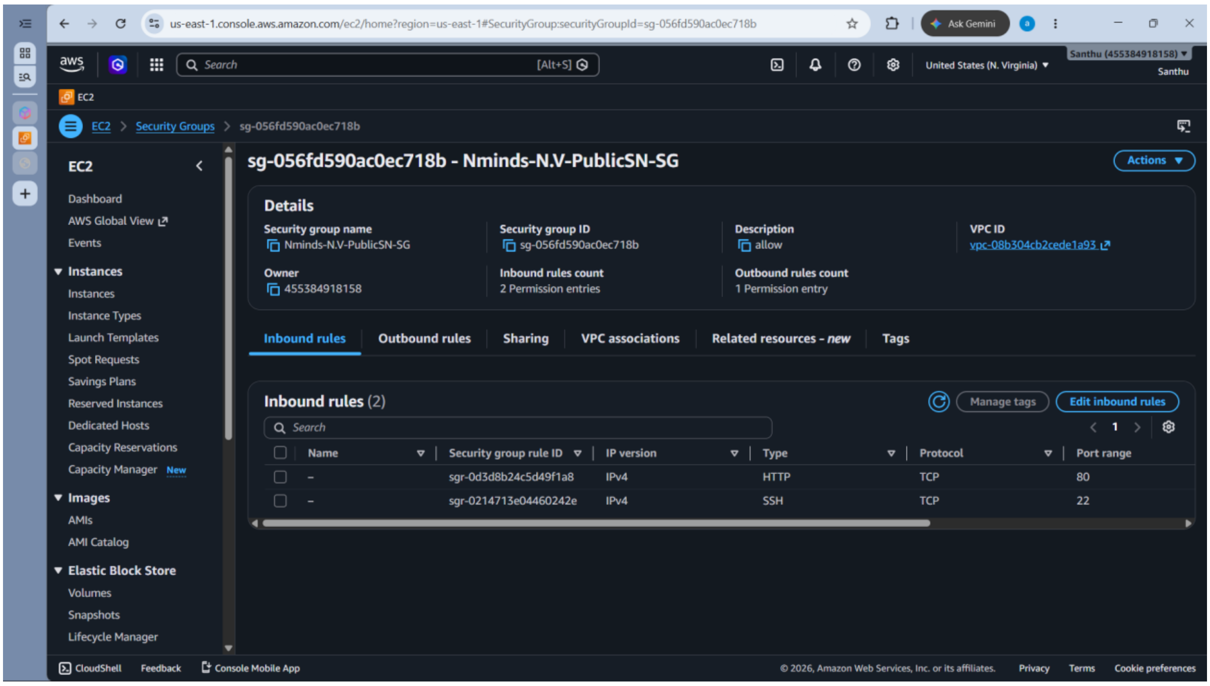

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#InstanceDetails:instanceId=i-02a5124c8df0e8ee6
EC2 > Instances > i-02a5124c8df0e8ee6
Instance summary for i-02a5124c8df0e8ee6 (Nminds-PublicSN-Instance) Info
Connect
Instance state
Actions
Updated less than a minute ago
Instance ID
i-02a5124c8df0e8ee6
Public IPv4 address
44.203.216.108 | open address
Private IPv4 addresses
10.0.100.254
IPv6 address
Instance state
Running
Public DNS
Hostname type
IP name: ip-10-0-100-254.ec2.internal
Private IP DNS name (IPv4 only)
ip-10-0-100-254.ec2.internal
Answer private resource DNS name
Instance type
t3.micro
Elastic IP addresses
Auto-assigned IP address
44.203.216.108 [Public IP]
VPC ID
vpc-0e1c16f5e275340d9 (Nminds-N.V-VPC)
AWS Compute Optimizer finding
IAM role
Subnet ID
subnet-04017452448aa88ee (Nminds-PublicSN)
Auto Scaling Group name
IMDSv2
Required
Instance ARN
arn:aws:ec2:us-east-1:455384918158:instance/i-02a5124c8df0e8ee6
Managed
false
Operator

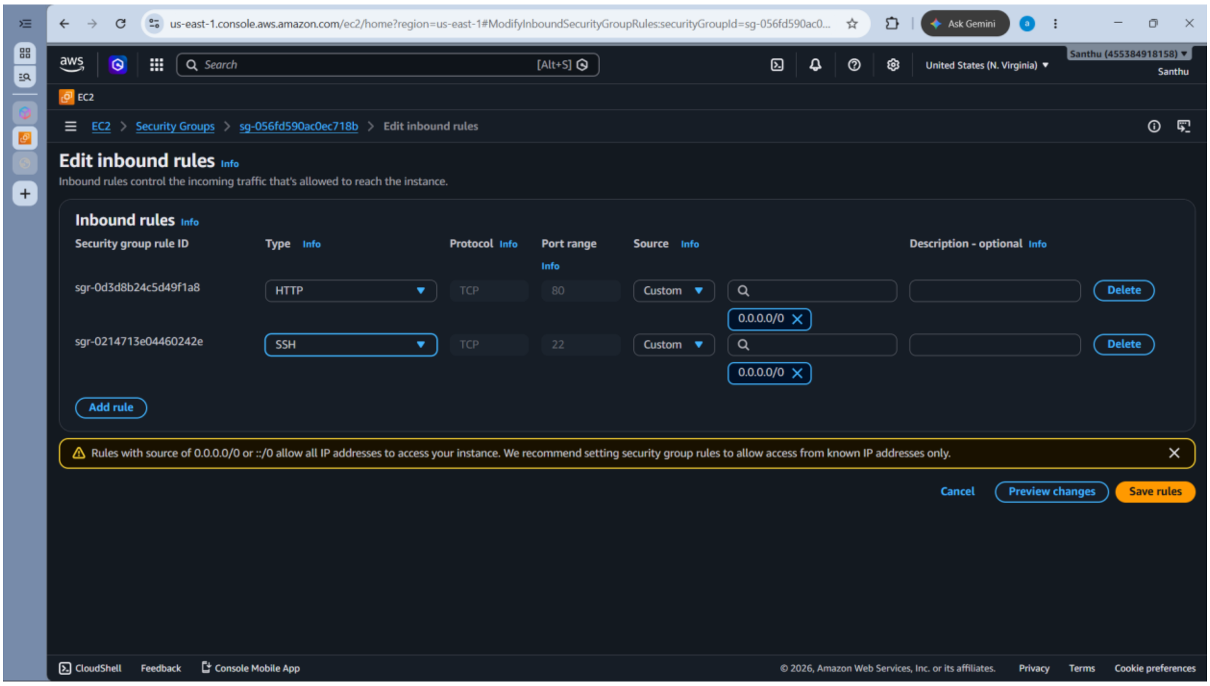
Task2: Create Apache Webserver (EC2) on newly created public subnet:

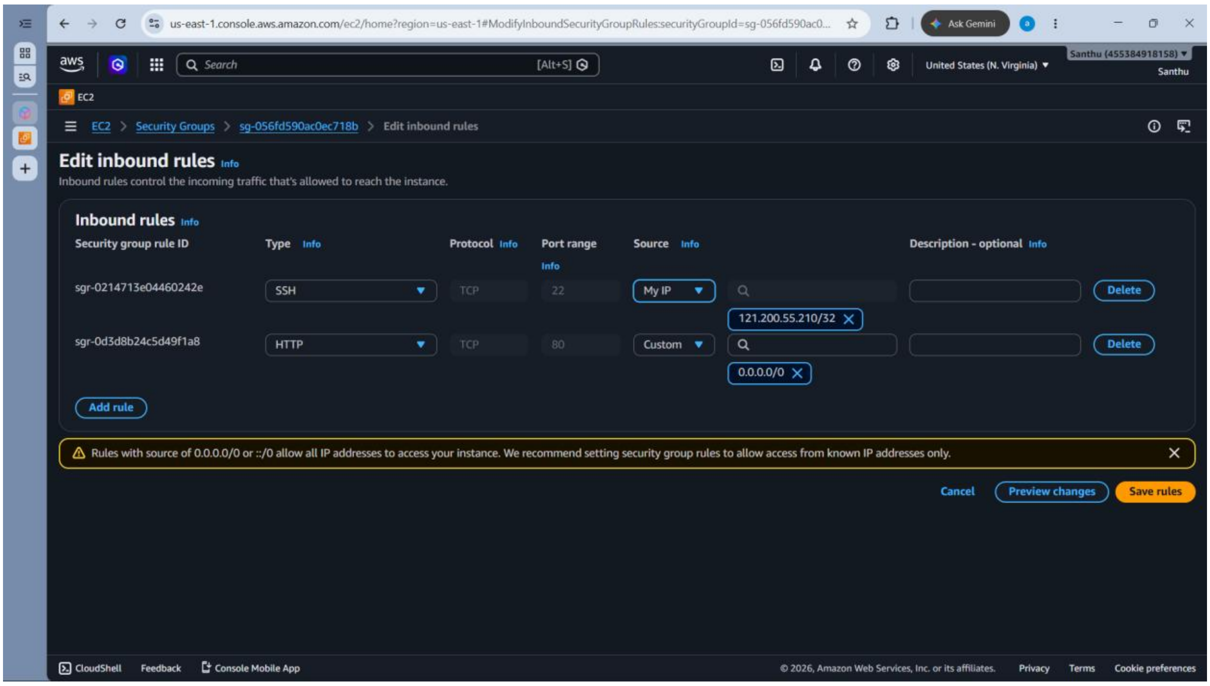




Task3: The Linux machine only able to access by system admin, how to configure for this scenario with execution steps







us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#ModifyInboundSecurityGroupRules:securityGroupId=sg-056fd590ac0...
EC2 > Security Groups > sg-056fd590ac0ec718b > Edit inbound rules
Edit inbound rules
Inbound rules control the incoming traffic that's allowed to reach the instance.
Inbound rules
Security group rule ID
Type
Protocol
Port range
Source
Description - optional
sgr-0214713e04460242e
SSH
TCP
22
My IP
121.200.55.210/32
Delete
sgr-0d3d8b24c5d49f1a8
HTTP
TCP
80
Custom
0.0.0.0/0
Delete
Add rule
Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.
Cancel
Preview changes
Save rules